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1a)

For implication introduction:

From $\neg A \vee B$ we can derive $A \rightarrow B$

$\neg A \vee B \vdash A \rightarrow B$

A	B	$\neg A \vee B$	$A \rightarrow B$
0	0	1	1
0	1	1	1
1	0	0	0
1	1	1	1

All models that satisfy $A \rightarrow B$ satisfy device sentence $\neg A \vee B$ so $\neg A \vee B$ is entailed so implication elimination is sound rule of inference

1b.

A	B	C	D	$A \wedge B$	$C \wedge D$	$A \wedge B \rightarrow C \wedge D$	$A \wedge B \rightarrow C$
0	0	0	0	0	0	1	1
0	0	0	1	0	0	1	1
0	0	1	0	0	0	1	1
0	1	0	0	0	0	1	1
1	0	0	0	0	0	1	1
1	1	0	0	1	0	0	0
1	0	1	0	0	0	1	1
1	0	0	1	0	0	1	1
0	1	1	0	0	0	1	1
0	1	0	1	0	0	1	1
0	0	1	1	0	1	1	1
1	1	1	0	1	0	0	1
1	1	0	1	1	0	0	0
1	0	1	1	0	1	1	1
0	1	1	1	0	1	1	1
1	1	1	1	1	1	1	1

All the models that satisfy $A \wedge B \rightarrow C \wedge D$ also satisfy $A \wedge B \rightarrow C$ so $A \wedge B \rightarrow C$ is entailed so $A \wedge B \rightarrow C$ is sound rule of inference

1c. Natural Deduction to show $A \wedge B \rightarrow C \wedge D \models A \wedge B \rightarrow C$

Premises

1. $A \wedge B \rightarrow C \wedge D$

Derivatives

2. $\neg(A \wedge B) \vee (C \wedge D)$ [Implication Elimination, 1]
3. $\neg A \vee \neg B \vee (C \wedge D)$ [DeMorgan's Law, 2]
4. $(\neg A \vee \neg B \vee C) \wedge (\neg A \vee \neg B \vee D)$ [Distributivity, 3]
5. $\neg A \vee \neg B \vee C$ [And Elimination]
6. $\neg A \vee C \vee \neg B$
7. $A \wedge B \rightarrow C$ [Horn clause(1a), 6]

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1d. Prove $A \wedge B \rightarrow C \wedge D \models A \wedge B \rightarrow C$ using Resolution

KB :

1. $\{A \wedge B \rightarrow C \wedge D\}$

Query: $A \wedge B \rightarrow C$

CNF:

1. $\neg A \vee \neg B \vee (C \wedge D)$
2. $A \wedge B \wedge \neg C$ // negated query

Reso

3. $\neg A \vee \neg B \vee C$ [AndElimination]
4. A [AndElimination, 2]
5. B [AndElimination, 2]
6. $\neg C$ [AndElimination, 2]
7. $\neg B \vee C$ [Reso, 3, 4]
8. C [Reso, 5, 7]
9. [Reso, 6, 8]; empty clause!

2a)

1. $C1W \rightarrow \neg L1W$
2. $C2W \rightarrow \neg L2W$
3. $C3W \rightarrow \neg L3W$
4. $C1Y \rightarrow \neg L1Y$
5. $C2Y \rightarrow \neg L2Y$
6. $C3Y \rightarrow \neg L3Y$
7. $C1W \rightarrow L1B \sqcup L1Y$
8. $C2W \rightarrow L2B \sqcup L2Y$
9. $C3W \rightarrow L3B \sqcup L3Y$
10. $C1Y \rightarrow L1B \sqcup L1W$
11. $C2Y \rightarrow L2B \sqcup L2W$
12. $C3Y \rightarrow L3B \sqcup L3W$
13. $C1W \rightarrow O1W$
14. $C2W \rightarrow O2W$
15. $C3W \rightarrow O3W$
16. $C1Y \rightarrow O1Y$
17. $C2Y \rightarrow O2Y$
18. $C3Y \rightarrow O3Y$
19. $C1B \rightarrow O1W \sqcup O1Y$
20. $C2B \rightarrow O2W \sqcup O2Y$
21. $C3B \rightarrow O3W \sqcup O3Y$
22. $C1B \rightarrow \neg L1B$
23. $C2B \rightarrow \neg L2B$
24. $C3B \rightarrow \neg L3B$
25. $C1B \rightarrow L1Y \sqcup L1W$
26. $C2B \rightarrow L2Y \sqcup L2W$
27. $C3B \rightarrow L3Y \sqcup L3W$
28. $\neg C1B \wedge \neg C1W \rightarrow O1Y$
29. $\neg C2B \wedge \neg C2W \rightarrow O2Y$
30. $\neg C3B \wedge \neg C3W \rightarrow O3Y$
31. $\neg C1B \wedge \neg C1Y \rightarrow O1W$
32. $\neg C2B \wedge \neg C2Y \rightarrow O2W$
33. $\neg C3B \wedge \neg C3Y \rightarrow O3W$
34. $C1W \rightarrow O1W \wedge \neg L1W$
35. $C2W \rightarrow O2W \wedge \neg L2W$
36. $C3W \rightarrow O3W \wedge \neg L3W$
37. $C1Y \rightarrow O1Y \wedge \neg L1Y$
38. $C2Y \rightarrow O2Y \wedge \neg L2Y$
39. $C3Y \rightarrow O3Y \wedge \neg L3Y$
40. $\neg C1W \wedge \neg C2B \rightarrow C3W$

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- 41. $\neg C2W \wedge \neg C3B \rightarrow C1W$
- 42. $\neg C3W \wedge \neg C1B \rightarrow C2W$
- 43. $\neg C1W \wedge \neg C3B \rightarrow C2W$
- 44. $\neg C1Y \wedge \neg C2B \rightarrow C3Y$
- 45. $\neg C2Y \wedge \neg C3B \rightarrow C1Y$
- 46. $\neg C3Y \wedge \neg C1B \rightarrow C2Y$

2b.

Facts = {46. O1Y, 47. L1W, 48. O2W, 49. L2Y, 50. O3Y, 51. L3B}

- 52. $\neg C1W \vee (O1W \wedge \neg L1W)$ [IE, 34]
- 53. $(\neg C1W \vee O1W) \wedge (\neg C1W \vee \neg L1W)$ [Distributivity, 52]
- 54. $\neg L1W \vee \neg C1W$ [AE, 54]
- 55. $L1W \rightarrow \neg C1W$ [ImplicationIntroduction, 54]
- 56. $\neg C1W$ [MP, 55, 47]
- 57. $\neg C3B \vee \neg L3B$ [IE, 24]
- 58. $\neg L3B \vee \neg C3B$ [Commutativity, 57]
- 59. $L3B \rightarrow \neg C3B$ [ImplicationIntroduction, 58]
- 60. $\neg C3B$ [MP, 51, 59]
- 61. $\neg C1W \wedge \neg C3B$ [AndIntroduction]
- 62. $C2W$ [MP, 61, 43]

2c.

CNF

1. $\neg C1W \sqcap \neg L1W$
2. $\neg C2W \sqcap \neg L2W$
3. $\neg C3W \sqcap \neg L3W$
4. $\neg C1Y \sqcap \neg L1Y$
5. $\neg C2Y \sqcap \neg L2Y$
6. $\neg C3Y \sqcap \neg L3Y$
7. $\neg C1W \sqcap L1B \sqcap L1Y$
8. $\neg C2W \sqcap L2B \sqcap L2Y$
9. $\neg C3W \sqcap L3B \sqcap L3Y$
10. $\neg C1Y \sqcap L1B \sqcap L1W$
11. $\neg C2Y \sqcap L2B \sqcap L2W$
12. $\neg C3Y \sqcap L3B \sqcap L3W$
13. $\neg C1W \sqcap O1W$
14. $\neg C2W \sqcap O2W$
15. $\neg C3W \sqcap O3W$
16. $\neg C1Y \sqcap O1Y$
17. $\neg C2Y \sqcap O2Y$
18. $\neg C3Y \sqcap O3Y$
19. $\neg C1B \sqcap O1W \sqcap O1Y$
20. $\neg C2B \sqcap O2W \sqcap O2Y$
21. $\neg C3B \sqcap O3W \sqcap O3Y$
22. $\neg C1B \sqcap \neg L1B$
23. $\neg C2B \sqcap \neg L2B$
24. $\neg C3B \sqcap \neg L3B$
25. $\neg C1B \sqcap L1Y \sqcap L1W$
26. $\neg C2B \sqcap L2Y \sqcap L2W$
27. $\neg C3B \sqcap L3Y \sqcap L3W$
28. $C1B \sqcap C1W \sqcap O1Y$
29. $C2B \sqcap C2W \sqcap O2Y$
30. $C3B \sqcap C3W \sqcap O3Y$
31. $C1B \sqcap C1Y \sqcap O1W$
32. $C2B \sqcap C2Y \sqcap O2W$
33. $C3B \sqcap C3Y \sqcap O3W$
34. $\neg C1W \vee O1W$
35. $\neg C1W \vee \neg L1W$

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- 36. $\neg C2W \vee O2W$
- 37. $\neg C2W \vee \neg L2W$
- 38. $\neg C3W \vee O3W$
- 39. $\neg C3W \vee \neg L3W$
- 40. $\neg C1Y \vee O1Y$
- 41. $\neg C1Y \vee \neg L1Y$
- 42. $\neg C2Y \vee O2Y$
- 43. $\neg C2Y \vee \neg L2Y$
- 44. $\neg C3Y \vee O3Y$
- 45. $\neg C3Y \vee \neg L3Y$
- 46. $C1W \vee C2B \vee C3W$
- 47. $C2W \vee C3B \vee C1W$
- 48. $C3W \vee C1B \vee C2W$
- 49. $C1Y \vee \neg C2B \vee C3Y$
- 50. $C2Y \vee C3B \vee C1Y$
- 51. $C3Y \vee C1B \vee C2Y$

2d.

Facts = {52. $O1Y$, 53. $L1W$, 54. $O2W$, 55. $L2Y$, 56. $O3Y$, 57. $L3B$ 58. $\neg C2W$ // negation of query}

New Clauses	Annotation
59. $\neg C3B$	Reso on 57 and 24($\neg C3B \square \neg L3B$)
60. $C2W \vee C1W$	Reso on 59 and 47
61. $\neg C1W$	Reso on 52 and 16
62. $C2W$	Reso on 60 and 61
63. Proof succeeds. Derived empty clause	Reso on 58 and 62

3.

Rule e is triggered because HaveMountainBike which causes HaveBike to be pushed into agenda

Rule m is triggered because AvisOpen which causes CarRentalOpen to be pushed into agenda

Rule o is then triggered because CarRentalOpen has been inferred meaning IsNotAHoliday is pushed into agenda

Rule k is triggered because HaveMoney and CarRentalOpen which infers CanRentCar

Rule j is then triggered because of CanRentCar which infers CanDriveToWork

Rule b is then triggered because CanDriveToWork which infers CanGetToWork

We then have found what we are looking for showing query is entailed and agenda becomes empty

The inferred positions are HaveBike, CarRentalOpen, and IsNotAHoliday, CanRentCar, CanDriveToWork, CanGetToWork.

CanGetToWork is among the inferred positions

4.

Goal stack	
CanGetToWork	Push query
CanBikeToWork	Rule a
<u>HaveBike WorkCloseToHome Sunny</u>	Try rule d
<u>HaveMountainBike WorkCloseToHome Sunny</u>	Try rule e for HaveBike
<u>WorkCloseToHome Sunny</u>	Succeeds. HaveMountainBike pops off
<u>Sunny</u>	Succeeds. WorkCloseToHome pops off
	Since Sunny is not provable. Backtrack to another for CanGetToWork.
CanDriveToWork	Try rule b
OwnCar	
	Since OwnCar isn't provable. BackTrack to another for CanDriveToWork
CanRentCar	Try rule j
<u>HaveMoney CarRentalOpen</u>	Try rule k
<u>CarRentalOpen</u>	Succeeds. HaveMoney pops off
HertzOpen	Since HertzOpen isn't provable or is false. Backtrack to another rule for Car Rental Open
AvisOpen	Succeeds. Pop off Avis Open
	Stack becomes empty. Proof succeeds.